

Review: "Prompt Engineering Techniques: Streamlining AI Workflows for Academic Deadlines" by Gauri Shirke

This article tackles prompt engineering through the relatable problem of a stressed college student trying to get useful help from an AI assistant at 2:30 AM. Rather than opening with dry technical definitions, the piece frames prompting as something we already do naturally the same way we rephrase a question for a friend to be better understood and then builds outward into a structured toolkit.

The article breaks prompting down into five core building blocks (role, task, instructions, context, input) before moving through techniques of increasing complexity: zero-shot, one-shot, and few-shot prompting for quick or pattern-based requests, followed by advanced methods like Chain of Thought, Role-based, Contextual, Retrieval-Augmented, and Tree of Thought prompting. Each technique comes with a practical, assignment-related example which was a nice touch — it bridges the conceptual explanation with something actually usable.

The two techniques that stood out most to me were Chain of Thought and Contextual prompting, mainly because I end up using both almost every time I work with AI. Chain of Thought is my go-to whenever I need to actually think through a problem instead of getting a final answer handed to me forcing the AI to lay out its reasoning step by step makes it way easier to spot where the logic breaks or where I need to redirect it. Generally, I use it whenever I am trying to learn something through a problem of DSA or CP. Contextual prompting is the one I probably rely on even more, honestly, since without background details the responses always feel generic. The moment I add context like deadlines, constraints, what I've already tried, the answers get noticeably sharper. Reading this made me realize I'd been doing this instinctively without knowing there was a name for it. I also use this for keeping a track on my progress in whichever field I want to.

This connects to broader conversations around AI literacy right now as more students rely on LLMs daily. Knowing how to structure a request is becoming as important as knowing what to ask. The article's closing insight that refining how you prompt AI also sharpens how you communicate with people is a subtle but effective takeaway that elevates it beyond a simple technique list.